TLS-SE 3.03
APDU INTERFACE

CONTENT

1 The TLS-SE Strict Application

This application supports only three APDU commands RECV, SEND, and SELECT (optional).

1.1 The Select Command

This command starts the TLS-SE application
Upon success it returns the status word SW1 SW2 = 9000

Command

CLA	INS	P1	P2	P3	AID
00	A4	04	00	06	010203040500

Response

SW1	SW2
90	00

1.2 The SEND Command

This command writes a TLS packet to the secure element, and returns a TLS message.
It requires no PIN.

Command

CLA	INS	P1	P2	P3	Data
00	D8	0 Normal Operation 1 if tls is open: Decrypt 2 if tls is open: Encrypt 3 if tls is open: Echo FF Rx Echo 1 in ClientHello PSK test 81 in ClientHelloPKI test	0:more 1: first 2: last 3: first&last	Data Length 00: Start/Reset TLS session	Data to be written

Response

Body	SW1	SW2	Comment
Optional message	90	00	OK
Optional message	90	01	TLS channel is opened
Optional message	90	02	TLS channel is closed
Optional message	61	length	More data (length bytes) to read
Optional message	9F	length	More data (length bytes) to read
Empty	6D	02	Write Error
Empty	6D	01	Read Error
Empty	6D	14	Decryption Error
Empty	6D	15	Encryption Error
Empty	6D	16	TLS Error

1.3 The RECV Command

This command reads a TLS packet
It requires no PIN.

Command

CLA	INS	P1	P2	P3
00	C0	0	0	Data Length

Response

Body	SW1	SW2	Comment
TLS Packet	90	00	No more date to read
TLS Packet	61	Length	More data (length bytes) to read
TLS Packet	9F	Length	More data (length bytes) to read

2 The TLS-SE Combi Application

This application is dedicated for testing purposes, and is not a strict TLS-SE application, which supports only three commands RECV, SEND, and SELECT (optional). It provides debug facilities and TLS-IM features.

2.1 The Select Command

This command starts the TLS-SE application
Upon success it returns the status word SW1 SW2 = 9000

Command

CLA	INS	P1	P2	P3	AID
00	A4	04	00	06	010203040500

Response

SW1	SW2
90	00

2.2 The Verify UserPin Command

This command verifies the user pin. The UserPin is required for the signature operations.

Upon success it returns the status word SW1 SW2 = 9000
Otherwise it returns SW1=63, SW2=number of remaining tries (3 at the most)

Command

CLA	INS	P1	P2	P3	UserPin
00	20	00	00	04	3030303030

Response

SW1	SW2	Comment
90	00	Success
63	Number of remaining tries	Fail
67	00	Wrong Length
6B	00	Wrong P1P2

2.3 The Verify AdminPin command

This command verifies the administrator pin. It gives access to all available features of the crypto currency application. If P2 is set to FF UserPin is reset to the default value (four zeros).

Upon success it returns the status word SW1 SW2 = 9000
Otherwise it returns SW1=63, SW2=number of remaining tries (ten at the most)

Command

CLA	INS	P1	P2	P3	AdminPin
00	20	00	01 FF Reset to default	08	3030303030303030

Response

SW1	SW2	Comment
90	00	Success
63	Number of remaining tries	Fail
67	00	Wrong Length
6B	00	Wrong P1P2

2.4 The ChangePin command

This command sets a PIN (UserPin,, AdminPin) to a new value
The P2 value is respectively 00, 01 for UserPin andAdminPin.
Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=number of remaining tries.

Command

CLA	INS	P1	P2	P3	OldPin	NewPIN
00	24	00	00	10	30303030FFFFFFFF	31313131FFFFFFFF
00	24	00	01	10	3030303030303030	3131313131313131

Response

SW1	SW2	Comment
90	00	Success
63	Number of remaining tries	Fail
67	00	Wrong Length
6B	00	Wrong P1P2

2.5 The GetStatus command

This command returns the current state of the crypto currency application. It required at least the previous checking of one PIN (UserPin, AdminPin).

Command

CLA	INS	P1	P2	P3
00	87	00	00	0A

Response

SW1	SW2	Comment
90	00	Success
63	80	PIN required

This command returns 8 bytes.

B0 B1 B2 B3: version= 3.0.3
B4 B5: RAM_SIZE MSB RAM_SIZE LSB (initial)
B6 B7: RAM_SIZE MSB RAM_SIZE LSB (remaing)

2.6 The Write Command

This command writes data in the non volatile memory, in the area [0000, 13FF] (5120 Bytes)

It requires the previous checking of PINs UserPin or AdminPin.
Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3	Data
00	D0	AdrMSB	AdrLSB	Data Length	Data to be written

The starting address ranging from 0000 to 13FF is encoded by two bytes (P1, P2), P1 being the *most significant byte* (MSB) and P2 the *less significant byte* (LSB).

Address Mapping

Start Address	Length	Comment	PIN required
0000	0800	Data Area	User or Admin

Response

SW1	SW2	comment
90	00	OK
63	80	PIN required
6D	02	Invalid Address

2.7 The Read Command

This command reads data in the non volatile memory.
It requires the previous checking of PINs UserPin or AdminPin.
Upon success it returns the status word SW1,SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3
00	B0	AdrMSB	AdrLSB	Length to be read

The starting address ranging from 0000 to 13FF (5120 bytes) is encoded by two bytes (P1, P2), P1 being the most significant byte (MSB) and P2 the less significant byte (LSB).

Address Mapping

Start Address	Length	Comment	PIN required
0000	0800	Data Area	User or Admin

Response

Body	SW1	SW2	comment
Data	90	00	OK
Empty	63	80	PIN required
Empy	6D	01	Invalid Address

2.8 The Clear KeyPair command

This command MUST be used before any key setting or key generation operation.
This command clears the curve parameters.
The InitCurve command is required to configure the Elliptic Curve
It clears RSA20248 key or EC256 key.

It requires the Admin PIN.

Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3	PIN required
00	81	00	AlgoId b5 b5=0 EC256, b5=1 RSA2048 Key Index b1b0	00	Admin

Response

SW1	SW2	Comment
90	00	Key Reset Done
63	80	PIN required
69	85	Bad index

2.9 The InitCurve command

This command initializes by default the elliptic curve SECP256r1 (P256) parameters, or SECP256k1

The keys MUST be cleared before this operation.

It requires the Admin PIN.

Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3	PIN required
00	89	FF= SECP256k1 other=SECP256 prime	Key Index [0,3]	00	Admin

Response

Data	SW1	SW2	Comment
	90	00	Key Reset Done
	63	80	PIN required
	69	85	Bad index
	64	01	Public Key is defined
	64	02	Private Key is defined
	6A	86	Incorrect P1P2

2.10 The Generate KeyPair Command

The keys MUST be cleared before this operation.

It requires the Admin PIN

Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3	PIN required
00	82	00	b5 AlgoId b5=0 EC256, b5=1 RSA2048 b1b0 :Key Index [0,3]	00	Admin

Response

SW1	SW2	Comment
90	00	OK
63	80	PIN required
69	85	Bad index
64	01	Public Key is defined
64	02	Private Key is defined
6D	10	Key Generation Error

2.11 The Get KeyParameter Command

This command collects the public and private keys parameters for EC256 or RSA 2048

It requires the User or the Admin PIN.
Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command for EC256

CLA	INS	P1	P2	P3	PIN required
00	84	00 Parameter A	b5=0, EC256 b1b0 :Key Index Ephemeral Key FF Cert Key FE	00	Admin or User
		01 Parameter B		00	Admin or User
		02 Parameter Field (Z/pZ)		00	Admin or User
		03 Parameter G (generator)		00	Admin or User
		04 Parameter K (cofactor)		00	Admin or User
		05 Parameter R (order of G)		00	Admin or User
		06 Parameter W (Public Key)		43	Admin or User
		07 Parameter S (Private Key		22	Admin Forbidden for KeyIndex FE

Response for EC256

Body	SW1	SW2	Comment
Param0 Length – Param0 value Param1 Length – Param1 value Param2 Length – Param2 value Param3 Length – Param3 value Param4 Length – Param4 value Param5 Length – Param5 value Param6 Length – Param6 value Param7 Length – Param7 value	90	00	Response includes a 2 bytes length field
	63	80	PIN required
	69	85	Bad index
	64	01	Public Key is not defined
	64	02	Private Key is not defined
	6A	86	Incorrect P1P2
	6D	30	Javacard Exception
ParamA Length – ParamA value	90	00	OK

Command for RSA2048

CLA	INS	P1	P2	P3	PIN required
00	84	00 Modulo	b5=1, RSA2048 b1b0 :Key Index	00	Admin or User
		01 Public Exponent		00	Admin or User
		02 P		00	Admin or User
		03 Q		00	Admin
		04 DP1		00	Admin
		05 DQ1		00	Admin
		06 PQ		00	Admin

Response for EC256

Body	SW1	SW2	Comment
Param0 Length – Param0 value Param1 Length – Param1 value Param2 Length – Param2 value Param3 Length – Param3 value Param4 Length – Param4 value Param5 Length – Param5 value Param6 Length – Param6 value	90	00	Response includes a 2 bytes length field
	63	80	PIN required
	69	85	Bad index
	64	01	Public Key is not defined
	64	02	Private Key is not defined
	6A	86	Incorrect P1P2
	6D	30	Javacard Exception
ParamA Length – ParamA value	90	00	OK

2.12 The SetKeyParameter EC256 Command

This command sets the elliptic curve parameters, including public and private keys parameters It requires the Admin PIN excepted for Parameter 6 (Public Key) for which User or Admin PIN is required.

Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3	PIN required
00	88	00 Parameter A	Key Index [0,3]	Length	Admin
		01 Parameter B		Length	Admin
		02 Parameter Field (Z/pZ)		Length	Admin
		03 Parameter G (generator)		Length	Admin
		04 Parameter K (cofactor)		Length	Admin
		05 Parameter R (order of G)		Length	Admin
		06 Parameter W (Public Key) The public key is checked according to the private key, and is reset in case of error		41	Admin
		07 Parameter S (Private Key The private key MUST be initialized before setting the public key		20	Admin

Response

SW1	SW2	Comment
90	00	OK
63	80	PIN required
69	85	Bad index
64	01	Public Key is defined
64	02	Private Key is defined
6A	86	Incorrect P1P2
6D	40	Javacard Exception

2.13 The SetKeyParameter RSA2048 Command

This command sets the RSA public and private keys parameters
It requires the Admin PIN.

Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1 b7b6b5b4 = param b1b0= KeyIndex	P2 modulo MSB	P3	PIN required
00	8C	00 Modulo	0	Length= 255	Admin
		10 Public Exponent	0	Length= 5	Admin
		20 P	0	Length=128	Admin
		30 Q	0	Length=128	Admin
		40 DP1	0	Length=128	Admin
		50 DQ1	0	Length=128	Admin
		60 PQ	0	Length=128	Admin

Response

SW1	SW2	Comment
90	00	OK
63	80	PIN required
69	85	Bad index
64	01	Public Key is defined
64	02	Private Key is defined
6A	86	Incorrect P1P2
6D	40	Javacard Exception

2.14 The SignECDSA command

This command generates an ECDSA or PKCS#1 1.5 signature.

It requires the AdminPin or UserPin.

Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).

Command

CLA	INS	P1	P2	P3	Data	PIN required
00	80	00 Raw Signature	b5: AlgoId 0=ECDSA, 1=PKCS#1 b1b0 Key Index FE for APP Key	Length 20	Data to be signed	Admin or User
00	80	21 Signature with SHA256	Key Index [0,3] FE or 5 for Key Cert Only for ECDSA	Length	Data to be hashed and signed	Admin or User

Response

Body	SW1	SW2	Comment
Length (2 bytes) ASN.1 ECDSA Signature Encoding Or RSA 256 byte PKCS#1 Signature	90	00	OK
	63	80	PIN required
	69	85	Bad index
	64	01	Public Key is not defined
	64	02	Private Key is not defined
	6A	86	Incorrect P1P2
	6D	20	Signature Error

2.15 The ECDHE command

This command performs Diffie Hellman key exchange over 256 bits elliptic curve and optionally generates a pair of public/ private keys . It requires no PIN.

Command

CLA	INS	P1	P2	P3	Data
00	8A	0	FF generation of an ephemeral public key over theSECP256 prime curve	41	PublicKey
		0	0…3 key index	41	PublicKey

Response

Body	SW1	SW2	Comment
DH value, 32 bytes	90	00	OK
Empty	69	95	Conditions Not Satisfied
Empty	64	03	Private key not definede
Empty	64	04	Public key not defined

2.16 The HMAC Command

This command is used to compute keys used by TLS1.3
It requires Admin or User PIN.

Command

CLA	INS	P1	P2	P3	Data	PIN
00	85	00	02= Compute HMAC LengthKey [Key] LengthData [Data]	2+ LengthKey+ LengthData	32 bytes	Admin or User
		00 FF	0A= EXTRACT_EARLY P1=00 return ESK P1=FF return ESK, HSK, eBSK, rBSK, feBSK, frBSK	23	01 00 20 PSK (32 bytes)	Admin or User
		00 01	0B EXPAND_EARLY P1=0 "c e traffic" P1=1 "e exp master"	3+ DataLength	00 20 DataLength Data	Admin or User
		00	0C= HMAC_EBSK	Data Lentgh	Data	Admin or User
		00	0D=HMAC_RBSK	Data Length	Data	Admin or User
		00	0E=EXTRACT_HANDSHAKE (HMAC_HSK)	Data Length	Data (PubKey)	Admin or User
		00	01= ADD Mod N N as defined for SECP256k1	Data Length =32	Data	Admin or User
		00 FF xy	05= AESCCM Encrypt P1= 00 AD length=5 P1=FF no AD Other= AD Length	16 (KEY) 12 (IV) AD Length+ Data Length	AESKey IV AD Data	Admin or User
		00 FF xy	06= AESCCM Decrypt P1= 00 AD length=5 P1=FF no AD Other= AD Length	16 (KEY) 12 (IV) AD Length+ Data Length+ 16 (Tag)	AESKey IV AD Data Tag	Admin or User
		16	Compute SHA256 P1= number of SHA256 update	Data Length	Data	Admin or User

Response

Body	SW1	SW2	Comment
P2=02, 0A, 0B, 0C, 0D, 0E, 16: 32 bytes 6x32 for = EXTRACT_EARLY with P1=FF	90	00	No Error
P1= 01, 32 bytes	90	00	No Error
P1= 05 AD EncryptedData Tag	90	00	No Error
P1=06 Decrypted Data	90	00	No Error
Empty	6A	86	Wrong P1 P2

2.17 The GenerateRandom command

This command computes a set of random bytes. It requires no PIN.

Command

CLA	INS	P1	P2	P3	PIN required
00	8B	00	00	Length	NO

Response

Body= P3 bytes or empty	SW1	SW2	Comment
	90	00	No Error
	69	85	Error

2.18 The GetCertificate command

This command reads the card certificate.
It requires the Admin or the User PIN.
Upon success it returns the status word SW1, SW2 = 9000.
Otherwise it returns SW1=63, SW2=80 (PIN required).
The card certificate is the ECDSA SHA256 signature over the secp56k1 curve, of its public key (in full representation, i.e. 65 bytes) located at index 0. This certificate is the concatenation of two 32 bytes integers value r, s.

Command

CLA	INS	P1	P2	P3	PIN required
00	8E	00	00	00	Admin or User

Response

Body	SW1	SW2	Comment
64 bytes certificate	90	00	Two 32 bytes integers r,s
Empty	63	80	PIN required

2.19 The SetCertificate command

This command sets the card certificate.
It requires the Admin PIN.
Upon success it returns the status word SW1, SW2 = 9000.

Two working mode are supported. The certificate (64 bytes) is imported or the certificate is internally computed from an embedded key (auto signing)

Command

CLA	INS	P1	P2	P3	Payload	PIN required
00	8F	00	00	40	64 bytes certificate	Admin
00	8F	FF	KeyIndex	00	Empty	Admin

Response

SW1	SW2	Comment
90	00	Certificate loaded or generated
63	80	PIN required
69	84	Certificate Already Set
69	85	Wrong Certificate Length
64	03	Private Key is not set

2.20 The SEND Command

This command writes a TLS packet to the secure element, a returns a TLS message.
It requires no PIN.

Command

CLA	INS	P1	P2	P3	Data
00	D8	0 Normal Operation 1 if tls is open: Decrypt 2 if tls is open: Encrypt 3 if tls is open: Echo FF Rx Echo 1 in ClientHello PSK test 81 in ClientHelloPKI test	0:more 1: first 2: last 3: first&last	Data Length 00: Start/Reset TLS session	Data to be written

Response

Body	SW1	SW2	Comment
Optional message	90	00	OK
Optional message	90	01	TLS channel is opened
Optional message	90	02	TLS channel is closed
Optional message	61	length	More data (length bytes) to read
Optional message	9F	length	More data (length bytes) to read
Empty	6D	02	Write Error
Empty	6D	01	Read Error
Empty	6D	14	Decryption Error
Empty	6D	15	Encryption Error
Empty	6D	16	TLS Error

2.21 The RECV Command

This command reads a TLS packet
It requires no PIN.

Command

CLA	INS	P1	P2	P3
00	C0	0	0	Data Length

Response

Body	SW1	SW2	Comment
TLS Packet	90	00	No more date to read
TLS Packet	61	Length	More data (length bytes) to read
TLS Packet	9F	Length	More data (length bytes) to read

2.22 The TEST command

This command performs test operations.
It requires the Admin PIN.

Command

CLA	INS	P1	P2	P3	Data
00	C0	0	01 DH test	20	DiffieHellman
		0	02 Public Key test	41	PublicKey
		0	03 random test	20	Random value
		0	0A Server Name	length	ServerName
		0	FF Shell Call	length	Shell Command

Response

Body	SW1	SW2	Comment
Optional Body	90	00	OK
Empty	67	00	Wrong Length
Empty	6D	02	Error Write
Empty	63	80	PIN required
Empty	6A	86	Wrong P1 P2